IMPORTANT CHECKLIST

- When you start the exam, **write the start time on your solutions.**
- When you stop the exam, **write the end time on your solutions**.
- Write the **honor code pledge**.

Solutions without these three ingredients **will not be graded**.

EXAM POLICY SUMMARY

- You have **exactly one contiguous hour** (no breaks) to work the exam questions.
- You are allowed one hand-written cheat sheet of at most one side of one page.
- **Justify all your answers**. It is not enough to just have the right answer. You must also convince us that you got the right answer for the right reasons.
- The complete exam policy (as posted on Canvas) is in force.

Each of the three questions is worth an equal number of points.

SUBMISSION CHECKLIST

- You must scan your solutions as well as your cheat sheet.
- Upload them to GradeScope as you do on a regular homework.

Solutions without a scanned cheat sheet **will not be graded**.

The exam questions are on the next page.

Good luck!

Q. 1 (Coronavirus patients)**.** Patients arrive at the hospital according to a Poisson process at an average rate of 20 per hour. Each patient is infected with coronavirus with probability 0.8. A patient with coronavirus is asymptomatic with probability $\frac{1}{4}$. The infection status and symptoms of different patients are independent of each other and of the arrival times. We assume that only a patient who is infected with coronavirus can exhibit symptoms.

a. What is the distribution of the first time at which a patient arrives at the hospital who is showing symptoms? What is the expectation of this time?

b. What is the probability that by the time the first symptom-exhibiting patient enters the hospital, an infected but asymptomatic patient had already arrived previously?

c. Given that the first 3 patients are not exhibiting symptoms, what is the probability that at least one of these patients is infected by coronavirus?

Q. 2 (Stocking up)**.** This morning the supermarket receives a shipment of disinfectant wipes. The number of packages in the shipment is Poisson distributed with mean 4. By the time the store opens, three people are already standing in line. Due to shortages, it is store policy that each customer can purchase at most one package of disinfectant wipes.

a. What is the probability that the store received a big enough shipment so that each of the three customers in line can get one package of disinfectant wipes?

In fact, each of the three customers is only in need of disinfectant wipes with probability $\frac{1}{2}$, independently of each other and of the size of the shipment.

b. What is the expected number of packages of wipes the store sells to the first three customers?

Q. 3 (Social distancing)**.** The parking lot at Trader Joe's consists of a single line of parking spots that is 24ft long. The entrance to the store is located on one end of the line.

Each arriving customer parks at a spot that is uniformly distributed along this line. Different customers choose their parking spots independently. (We assume for simplicity that no collisions are possible, e.g., cars can double park in the same spot.) Because of social distancing rules, however, customers must line up to get into Trader Joe's while standing at least 6ft apart.

a. What is the probability that the first customer parks within 6ft of the entrance to the store? What is the expected distance the first customer parks to the entrance?

b. What is the mean square distance between the spots where the first two customers park?

c. What is the probability that the first two customers who arrive park at least 6ft apart (and therefore do not have to move in order to satisfy the social distancing requirement)?